Linear Algebra Notes

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Chapter 1

Abstract Concepts

1.1. Vector Spaces

Let $\mathbb{F}$ be a field. A *vector space* over $\mathbb{F}$ is a nonempty set V that is closed under addition and scalar multiplication, and equipped with two operations:

$$V \times V \longrightarrow V, \qquad (u, v) \longmapsto u + v,$$

called *vector addition*, and

$$\mathbb{F} \times V \longrightarrow V, \qquad (a, v) \longmapsto av,$$

called *scalar multiplication*.

These operations are required to satisfy the following properties for all vectors $u, v, w \in V$ and all scalars $a, b \in \mathbb{F}$.

1. Associativity: $(u + v) + w = u + (v + w)$.
2. Commutativity: $u + v = v + u$.
3. Identity: There is a vector $0_V \in V$ such that $v + 0_V = v$.
4. Inversion: There is a vector $-v \in V$ such that $v + (-v) = 0_V$.
5. Scalar multiplication is compatible with multiplication in $\mathbb{F}$: $a(bv) = (ab)v$.
6. The multiplicative identity of $\mathbb{F}$ acts trivially: $1_{\mathbb{F}}v = v$.
7. Scalar multiplication distributes over vector addition: $a(u + v) = au + av$.
8. Scalar addition distributes over scalar multiplication: $(a + b)v = av + bv$.

The elements of V are called *vectors*, and the elements of $\mathbb{F}$ are called *scalars*.

If there is a linear bijection $f : V \to \mathbb{R}^n$, then we call n the dimension of V, denoted by $\dim V = n$.

A subset $U \subseteq V$ is called a *subspace of* V if U is itself a vector space over $\mathbb{F}$.

Given a field $\mathbb{F}$, define

$$\mathbb{F}^n := \mathbb{F} \times \cdots \times \mathbb{F} = \{(x_1, x_2, \ldots, x_n) \mid x_i \in \mathbb{F}, i = 1, 2, \ldots, n\}.$$

For every element $a \in \mathbb{F}^n$, the coordinates of a are denoted by $a = (a_1, a_2, \ldots, a_n)$.

Let $V_1, \ldots, V_n \subseteq V$ be subspaces of V. The *sum* of $V_1, \ldots, V_n$ is defined as

$$V_1 + \cdots + V_n := \{v_1 + \cdots + v_n \mid v_i \in V_i, i = 1, \ldots, n\}.$$

If for every $i \in \{1, \ldots, n\}$, one has

$$V_i \cap (V_1 + \cdots + V_{i-1} + V_{i+1} + \cdots + V_n) = \{0\},$$

we call this sum *direct sum* of $V_1, \ldots, V_n$, denoted by

$$V_1 \oplus \cdots \oplus V_n := V_1 + \cdots + V_n.$$

Proposition 1.1.1.

$V_1 + \cdots + V_n$ is the smallest subspace of V containing $V_1, \ldots, V_n$.

Proof. Clearly, $V_1 + \cdots + V_n$ is a subspace and contains every V_i. Let $U \subseteq V$ be any subspace containing $V_1, \ldots, V_n$. Let $v_i \in V_i$ for all i. Since U is a subspace, we have $v_1 + \cdots + v_n \in U$. So $V_1 + \cdots + V_n \subseteq U$. □

Proposition 1.1.2.

If $v \in V_1 \oplus \cdots \oplus V_n$, then v can be written in only one way $v = v_1 + \cdots + v_n$, where $v_i \in V_i$ for all i.

Proof. Suppose $v = v_1 + \cdots + v_n = u_1 + \cdots + u_n$, where $u_i \in V_i$ for all i. Then

$$(u_1 - v_1) + \cdots + (u_n - v_n) = x_1 + \cdots + x_n = 0$$

where $x_i = u_i - v_i \in V_i$. One has

$$x_1 = -x_2 - \cdots - x_n.$$

Since the RHS vector belongs to $V_2 \oplus \cdots \oplus V_n$, $x_1 \in V_1$, and

$$V_1 \cap V_2 \oplus \cdots \oplus V_n = \{0\}.$$

Therefore, $x_1 \notin V_2 \oplus \cdots \oplus V_n$, contradiction. $\square$

Now let V be a vector space, $v_1, \ldots, v_n \in V$ be nonzero vectors. The *spanning space* of $v_1, \cdots, v_n$ is the vector space defined by

$$\langle v_1, \ldots, v_n \rangle = \{a_1 v_1 + \cdots + a_n v_n \mid (a_1, \ldots, a_n) \in \mathbb{F}^n\}.$$

We call $v_1, \cdots, v_n$ *linearly independent* if

$$a_1 v_1 + \cdots + a_n v_n = 0 \Leftrightarrow a_1 = \cdots = a_n = 0.$$

and *linearly dependent* if there exists $i < n$ and $a_i, \ldots, a_n \in \mathbb{F} \backslash \{0\}$ such that

$$a_i v_i + \cdots a_n v_n = 0$$

We call the set of vectors $\beta = \{v_1, \ldots, v_n\}$ a *basis of* V if $v_1, \ldots, v_n$ are linearly independent and $\langle v_1, \ldots, v_n \rangle = V$. The number n is called *the dimension of* V. Actually, every vector space has a basis, no matter if V is finite or infinite, which can be proved using Zorn's lemma. But we need a more general definition of linearly independent.

Let $U \subseteq V$ be any set, we call the set U *linearly independent* if for any $n \in \mathbb{N}$ and $u_1, \ldots, u_n \in U$, the vectors $u_1, \cdots, u_n$ are linearly independent.

We call the set U *a spanning set of* V if for every $v \in V$, there exists a sequence of (finite or countable) vectors $(v_k)_{k \in \mathbb{N}} \subseteq U$ and a corresponding sequence $(a_k)_{k \in \mathbb{N}} \subseteq \mathbb{F}^n$ such that

$$v = a_1 v_1 + a_2 v_2 + \cdots = \sum_{k=1}^{\infty} a_k v_k.$$

Lemma 1.1.3 (Zorn's Lemma).

Let $(\mathcal{P}, \leq)$ be a partially ordered set. Suppose that every chain $\mathcal{C} \subseteq \mathcal{P}$ has an upper bound in P. Then P contains a maximal element (meaning that there are no element greater than it).

The rigorous proof is complicated, see its elementary proof here if you want to be an IMO participant.

Theorem 1.1.4 (Hamel's Theorem).

Every vector space V has a basis.

Proof. Let

$$\mathcal{P} = \{S \subseteq V \mid S \text{ is linearly independent}\}.$$

We have $\varnothing \in \mathcal{P}$. Suppose $\mathcal{C} \subseteq \mathcal{P}$ is a chain of $\mathcal{P}$, let

$$P = \bigcup_{S \in \mathcal{C}} S.$$

Claim 1.1.5 (Proof of the claim). P is linearly independent, or $P \in \mathcal{P}$.

Proof. Let $u_1, \dots, u_m \in P$ and suppose

$$a_1 u_1 + \cdots + a_m u_m = 0.$$

Observe that for every i, there exists $S_i \in \mathcal{C}$ such that $u_i \in S_i$. Then one can assume that $S_1 \leq \cdots \leq S_m$. So $u_i \in S_m$ for all i. Hence $u_1, \cdots, u_m$ are linearly independent, which implies P is linearly independent. $\square$

By this claim, P is an upper bound of the claim $\mathcal{C}$. By the Zorn lemma, $\mathcal{P}$ attains a maximal element $\mathcal{M}$.

Claim 1.1.6. $\langle \mathcal{M} \rangle = V$.

Proof of the claim. Suppose $\langle \mathcal{M} \rangle \subsetneq V$. Let $v \in V \setminus \langle \mathcal{M} \rangle$ and $v \neq 0$. We will show that $\{v\} \cup \mathcal{M}$ is linearly independent. Indeed, let $u_1, \dots, u_m \in \mathcal{M}$. Suppose that

$$av + a_1 u_1 + \cdots + a_m u_m = 0$$

If $a \neq 0$, then

$$v = -\frac{a_1}{a} u_1 - \cdots - \frac{a_m}{a} u_m.$$

Hence, $v \in \langle M \rangle$, contradiction. So $a = 0$, then $a_1 u_1 + \cdots + a_m u_m = 0$. Since $u_1, \dots, u_m$ are linearly independent, we obtain $a = a_1 = \cdots = a_m = 0$. Then $\{v\} \cup \mathcal{M} \in \mathcal{P}$ and $\{v\} \cup \mathcal{M} \supsetneq \mathcal{M}$. This is also contradiction since $\mathcal{M}$ is a maximal element of $\mathcal{P}$. $\square$

This shows that $\mathcal{M}$ is a basis of V. $\square$

Theorem 1.1.7 (Steinitz's Exchange Lemma).

Corollary 1.1.8. Let V be a finite vector space and $U \subseteq V$ is a subspace. Then there exists a subspace $W \subseteq V$ such that $U \oplus W = V$.

Proposition 1.1.9.

If $f : \mathbb{R}^m \to \mathbb{R}^n$ is a linear bijection, then $m = n$.

Proof. Let $e_1, \ldots, e_m$ be standard bases of $\mathbb{R}^m$. Given $x \in \mathbb{R}^n$, one may check that $f(e_1), \ldots, f(e_m)$ form a basis of $\mathbb{R}^n$. Indeed, let $y = f^{-1}(x) = y_1 e_1 + \cdots + y_m e_m$, then

$$x = y_1 f(e_1) + \cdots + y_m f(e_m)$$

Thus $f(e_1), \ldots, f(e_m)$ span $\mathbb{R}^n$. Now suppose

$$y_1 f(e_1) + \cdots + y_m f(e_m) = 0.$$

One has

$$f(y_1 e_1 + \cdots + y_m e_m) = 0.$$

Since f is linear, we have

$$y_1 e_1 + \cdots + y_m e_m = 0.$$

Since $e_1, \ldots, e_m$ is linearly independent, $y_1 = \cdots = y_m = 0$. Therefore, $f(e_1), \ldots, f(e_m)$ form a basis of $\mathbb{R}^n$, which implies $m = n$. □

In an infinite dimension, two bases in a vector space also have the same cardinality.

Proposition 1.1.10.

Let V be infinite-dimensional, if $\mathcal{A}$ and $\mathcal{B}$ are two bases of V, then $|\mathcal{A}| = |\mathcal{B}|$ in cardinality.

Proof.

□

Chapter 2

ODE Modeling

2.1. Archery

In this section, we will model the trajectory of the arrow in an hunting scenario. We will assume that the arrow is shot from a bow at a certain angle and initial draw length. The goal is to determine the optimal angle and draw length to hit a target at a given distance.

2.1.1. Modeling

Consider the 2D plane with the position $(0, h_0)$ where the arrow is shot, and the target at position $(d_0, 0)$ with height h_d.